SinoGNSS®

ComNav Technology

User Guide

M20 X GNSS/INS Receiver



Rev 1.1

Revision history

Revision version	Description of Changes	Date
V1.0	Initial release	2025.12
V1.1	Updated COM2 commands and message description	2026.06

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Technical Assistance

If you have any question and can't find the answer in this manual, please contact your local dealer from which you purchased the M20 X High-precision integrated navigation receiver. Alternatively, request technical support from ComNav Technology Website: www.comnavtech.com or technical support email: support@comnavtech.com. Your feedback about this Guide will help us to improve it with future revisions.

Safety Information

Before using the receiver, please make sure that you have read and understood this User Guide, as well as the safety requirements.

- Connect your devices strictly based on this User Guide
- Install the GNSS receiver in a location that minimizes vibration and moisture
- Avoid falling to ground, or colliding with other items
- Keep a sound ventilation environment
- Change the cable if damaged

Related Regulations

The receiver contains integral Bluetooth wireless technology. Regulations regarding the use of the datalink vary greatly from country to country. In some countries, the unit can be used without obtaining an end-user license. But in some countries administrative permissions are required. For license information, please consult your local dealer.

Use and Care

The receiver can withstand the rough treatment that typically occurs in the field. However, the receiver is high-precision electronic equipment and should be treated with reasonable care.

Warning and Caution

An absence of specific alerts does not mean that there are no safety risks involved. A Warning or Caution information is intended to minimize the risk of personal injury and/or damage to the equipment.

WARNING- A Warning alerts you to a potential risk of serious injury to your person and/or damage to the equipment, because of improper operations or wrong settings of the equipment.

CAUTION- A Caution alerts you to a possible risk of damage to the equipment and/or data loss.

Warranty Notice

ComNav Technology does not warranty devices damage because of force majeure (lighting, high voltage or collision).

ComNav Technology does not warranty the disassembled devices.

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1 Product Overview

1.1 Product Appearance and Introduction

The M20X GNSS/INS receiver is shown in Figure 1-1.



Figure 1-1. M20X GNSS/INS receiver

The M20 X is a high-performance dual-antenna integrated navigation and positioning product that complies with automotive-grade design and manufacturing standards. It is based on MEMS inertial navigation technology and integrates satellite navigation (with RTK support) and vehicle information (such as wheel speed and gear signals).

By leveraging GNSS/INS/DR fusion algorithms, the M20 X delivers continuous high-precision position, velocity, and attitude data even in challenging environments such as urban canyons, overpasses, tree-lined roads, highways, and industrial parks where satellite signals may be blocked or affected by multipath.

It provides a reliable and stable high-precision positioning solution for applications such as autonomous driving, low-speed unmanned vehicles, smart ports, digital mines, trunk logistics, industrial park cleaning, ROBOTAXI, and ROBOBUS.

1.2 Product Features

- Proprietary core technology with a new-generation positioning and heading module, supporting multi-constellation, multi-frequency positioning and heading.

- Automotive-grade design with an IMU compliant with ASIL B requirements.
- Proprietary integrated navigation algorithm significantly improves positioning accuracy in challenging environments.
- Rich interfaces, including Automotive Ethernet, CAN, serial ports, and PPS, with support for external wheel-speed input.
- Supports high-dynamic applications and data output rates of up to 100 Hz.
- Compact and lightweight design with a magnesium-aluminum alloy enclosure.

1.3 Package List

Upon receiving the device, please check that all items in the package are complete. If any items are missing or damaged, please contact our staff promptly. The supplied items are listed in Table 1-1.

Table 1-1. Package list

Item Name	Quantity	Picture
M20 X GNSS/INS Receiver Main Unit	1 piece	
18 PIN Data Cable	1 piece	
FAKRA GNSS Antenna Cables (Black and Blue)	2 pieces	
High-Precision GNSS Antenna	2 pieces	

Item Name	Quantity	Picture
Strong Magnetic Mount	2 pieces	

1.4 Interface and Wiring Instructions

1.4.1 Interfaces

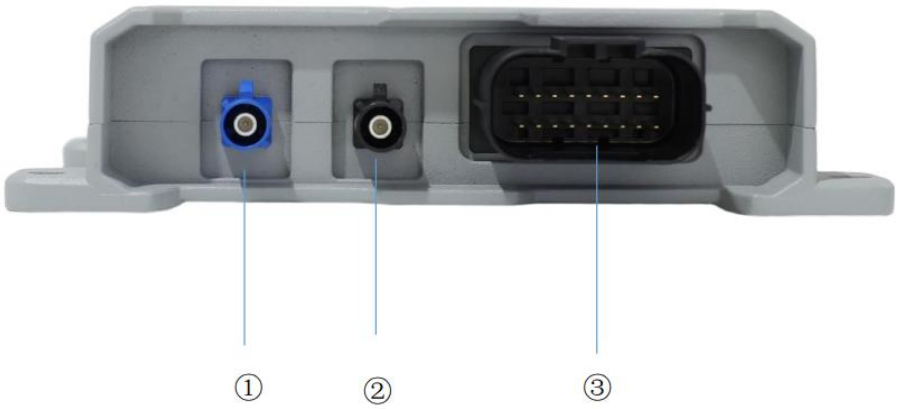


Figure 1-2. Receiver interfaces

① **GNSS2:** Secondary GNSS antenna interface, FAKRA-C

② **GNSS1:** Primary GNSS antenna interface, FAKRA-A

③ **DATA:** 18-pin data interface (DC + 2 × RS232 + Automotive Ethernet + PPS + 2 × CAN/CAN FD)

1.4.2 Wiring

Figure 1-3 shows the receiver connections. Figure 1-4 shows the wire harness and connector identifiers.

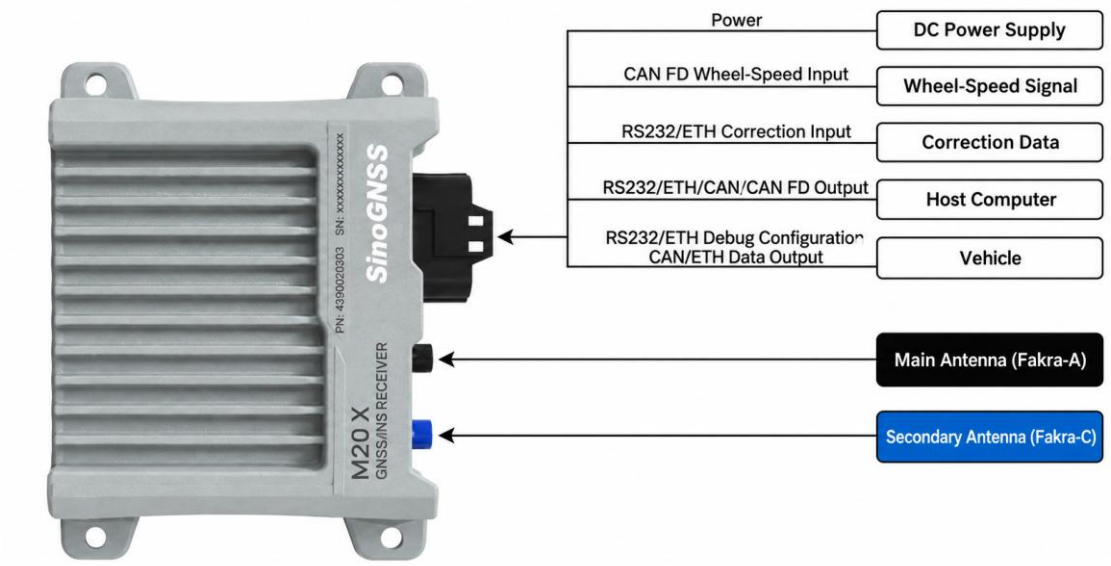


Figure 1-3. Receiver connection overview

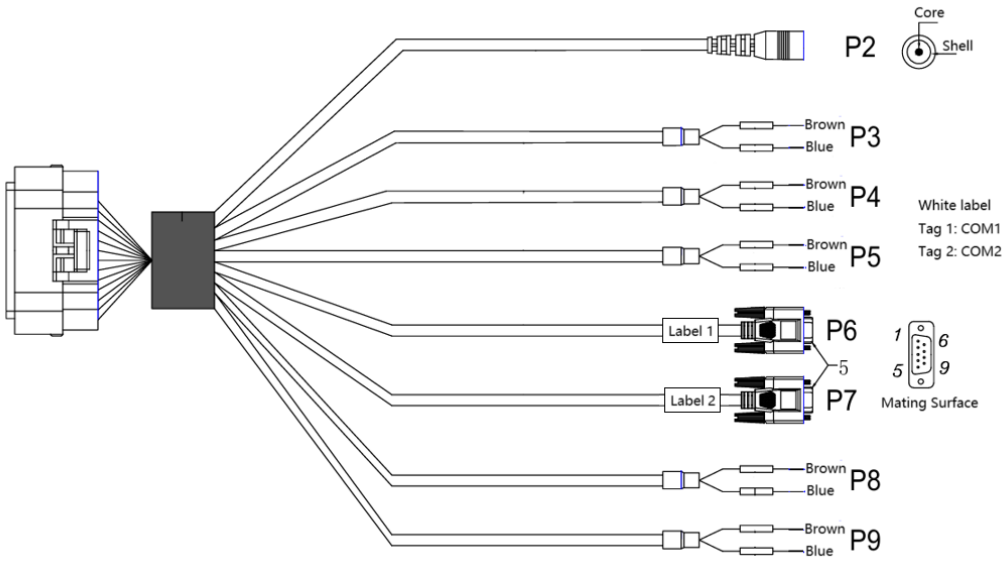


Figure 1-4. Wire harness and connectors

Table 1-2 defines the connector identifiers shown in Figure 1-4.

Table 1-2. Connector definitions

Pin	Definition		Description
1	P2		Power Supply

Pin	Definition		Description
2	P3	CAN1H	CAN1 High
		CAN1L	CAN1 Low
3	P4	CAN2H	CAN2 High
		CAN2L	CAN2 Low
4	P5	PPS	PPS 5V Signal
		GND	Power Ground
5	P6	COM1	Serial Port 1
6	P7	COM2	Serial Port 2
7	P8	ENET_P	Vehicle Ethernet Positive Data Line (P)
		ENET_N	Vehicle Ethernet Negative Data Line (N)
8	P9	RESERVED	Reserved
		GND	Power Ground

1.4.3 Accessory Connections

1. **Connect the GNSS antenna cables:**

   Connect the FAKRA connectors of the supplied FAKRA-A (black) and FAKRA-C (blue) antenna cables to the corresponding FAKRA RF sockets on the receiver. Make sure the connector colors are correctly matched.

2. **Connect the 18-pin data cable:**

   Align the latch on the 18-pin male connector of the data cable with the corresponding slot on the receiver's female socket, and then insert the connector fully.

3. **Connect the power supply:**

   After connecting the 18-pin data cable to the receiver, connect the DC barrel connector on the cable to an external power supply or battery. The receiver starts automatically when power is applied.

   Caution: Do not reverse the positive and negative terminals of the external power supply. The required input voltage is **5–32 V DC**.

4. **Connect the serial ports:**

   The M20X provides two serial ports, COM1 and COM2. Connect the DB9 female connector of the data cable to a computer or another communication device for command requests and integrated-navigation message transmission.

   Note: The baud rate of any external sensor must match the baud rate configured on the receiver.

5. **Connect the automotive Ethernet interface:**

   After connecting the 18-pin data cable to the receiver, connect the automotive Ethernet bare wires at the other end of the data/power cable to another automotive Ethernet device. To connect the receiver to a computer or other industrial Ethernet equipment, an automotive-to-industrial Ethernet converter is required.

The Ethernet interface supports IPv4 and TCP/UDP communication. It can be used for remote debugging, receiving correction data, outputting position and attitude information, and upgrading receiver firmware.

6. **Connect the CAN/CAN FD interface:**

   After connecting the 18-pin data cable to the receiver, connect the appropriate bare wires at the other end of the data/power cable to the user's synchronization device. Connect the GND wire to the ground of the user device.

The M20X provides a PPS signal for time synchronization, which may be used as required. By default, the receiver outputs a one-pulse-per-second PPS signal. The rising edge is synchronized with GPS time. When GNSS signal reception is good, the synchronization accuracy is **20 ns**.

7. **Connect the odometer or wheel-speed sensor:**

   The M20X supports connection to an odometer or wheel-speed sensor. Wheel-speed data can be transmitted to the M20X according to the DBC protocol. This information can improve integrated-navigation positioning accuracy during GNSS signal outages. Use this function according to the application requirements.

2 Installation Instructions

Before installing the M20X integrated navigation receiver, please read the installation instructions in this section carefully to avoid affecting the positioning results.

2.1 Installing the Antenna

First, install the GNSS antenna supplied in the kit onto the magnetic mounting base. Then rigidly mount the base onto the vehicle or carrier to ensure that the antenna does not shift or wobble while the carrier is moving.

The GNSS antenna should be installed horizontally. If it must be tilted, the tilt angle should not exceed 15°. Make sure the area above the GNSS antenna is open and free of obstructions.

Finally, securely screw the TNC connector of the RF cable onto the TNC connector of the antenna.

2.2 Installing the Receiver

The main unit should be securely and reliably mounted, with a rigid connection to the carrier, ensuring that the relative positions of the main unit, antenna, and carrier remain fixed. The following requirements apply to the installation area for the main unit:

1. The main unit should be installed on a rigid structure, with the surface flatness requirement of less than 1° (flatness refers to the pitch and roll angles between the rigid structure surface and the horizontal plane).
2. The temperature of interference units around the main unit should be kept below 85°C.
3. No vibration-interfering units should be within a 50 cm radius of the main unit. The unit should be placed away from vibration sources such as engines, subwoofers, and other areas prone to strong vibrations. The IMU should avoid the following frequency bands: **15.8kHz-17.8kHz, 28-35.6kHz, 44kHz, 18.3-20.3kHz, 1.9-2.1kHz**.
4. The main unit should be placed away from areas with complex electromagnetic environments, such as near LiDAR, instrument panels, and other equipment.

The recommended installation method is as follows, as shown in Figure 2-1:

1. Horizontal installation, with a horizontal angle of less than 5°, the smaller the better.
2. The X-axis of the main unit should point in the direction of the carrier’s forward motion, the Y-axis should be perpendicular to the forward direction and point to the right, and the Z-axis should point downward.
3. The X-axis of the main unit should be parallel to the vehicle's longitudinal centerline, with a distance of less than 70 cm.
4. The main unit should be installed at or near the center of the rear axle of the vehicle.

If the orientation of the main unit does not meet the above requirements, a horizontal deflection arrangement can be used with deflection angles in multiples of 90°.

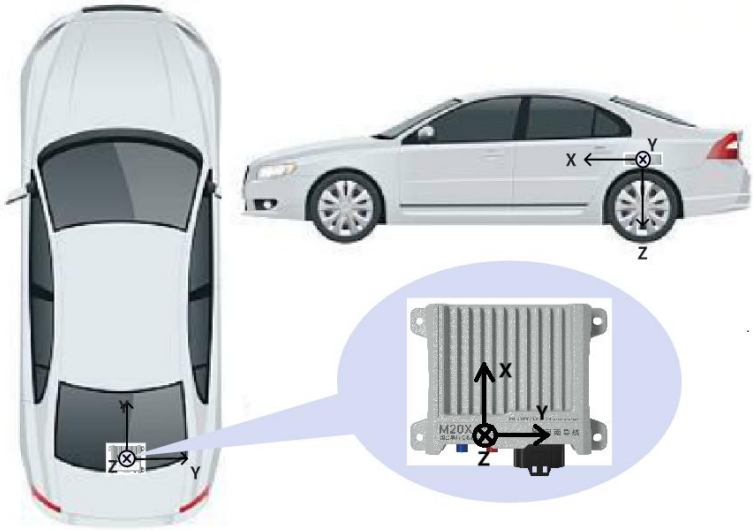


Figure 2-1. Recommended receiver installation

3 Receiver Configuration

The M20X integrated navigation receiver can be configured by sending commands via the serial port or Ethernet port.

3.1 User Configuration

3.1.1 Serial Port Configuration

The M20X can communicate with a computer or other devices through the serial port. Before communication is established, the serial port parameters on both devices must be configured properly. The default serial port settings are shown in Table 3-1.

Table 3-1. Default serial-port settings

Configuration Item	Value
Baud Rate	COM1:115200, COM2:115200 (default)
Parity	None
Data Bits	8 bit
Stop Bits	1 bit

Operating Procedure:

Use the following command to change the serial port configuration:

Set the baud rate of **COM1** to **115200**. Enter the command:

```
COM COM1 115200
```

Set the baud rate of **COM2** to **460800**. Enter the command:

```
SET COM2 460800
```

The M20X supports communication with a remote terminal through the serial port. Before establishing communication between the terminal software and the M20X, make sure that the serial port settings on both sides are consistent.

A computer can be used as the remote terminal for applications such as navigation data logging.

3.1.2 Installation Parameter Configuration

After the initial installation of the unit is completed, the relevant parameters need to be configured according to the installation position and installation method.

There are four mandatory configuration items:
1. GNSS antenna lever arm
2. Rear axle center lever arm
3. Unit rotation parameter configuration
4. Dual-antenna installation angle

For detailed definitions and configuration methods, refer to Sections 3.1.2.1 to 3.1.2.4.

Please note the following during the installation parameter configuration process:

1. After completing the configuration, send the **SAVECONFIG** command to save the parameters. Otherwise, the parameters will be lost after the receiver is powered off and will need to be configured again.
2. After saving the configuration parameters, send the **RESET** command or manually restart the receiver for the changes to take effect.
3. After configuration is completed, verify that the parameters have been saved correctly. After restarting the receiver, send the **READDRCONFIG** command and check the configuration information returned through the serial port to confirm that the parameters have been saved properly.
4. After reinstalling the receiver, the antenna lever arm, rear axle center lever arm, and receiver rotation parameters must be measured and configured again. If the GNSS antenna is relocated, the GNSS antenna lever arm and dual-antenna installation angle must also be remeasured and reconfigured. Before reconfiguring the receiver parameters, it is recommended to send the **CLEARDRCONFIG** command to clear the current configuration.
5. Manually measured receiver installation parameters are usually not sufficiently accurate. After completing the user configuration, it is recommended to use the online calibration function to estimate more accurate receiver installation parameters and improve the overall performance of the integrated navigation system.

3.1.2.1 GNSS Antenna Lever Arm Configuration

The vector from the origin of the receiver coordinate system to the phase center of the primary GNSS antenna is called the GNSS antenna lever arm. Use the receiver coordinate-system origin as the reference point and measure the X, Y, and Z coordinates of the primary antenna phase center along axes aligned with the vehicle coordinate system.

Configure the GNSS antenna lever arm with the following command:

```
SETLEVERARM ANT1 X Y Z
```

X, Y, and Z are the antenna lever-arm coordinates in meters. In Figure 3-1, X = 0.550 m, Y = −0.330 m, and Z = −0.670 m.

```
SETLEVERARM ANT1 0.550 -0.330 -0.670
```

Note: Antenna lever-arm errors directly affect the position accuracy of the integrated navigation solution. Keep the measurement error below 2 cm whenever possible. If accurate measurement is unavailable, run online calibration after configuration.

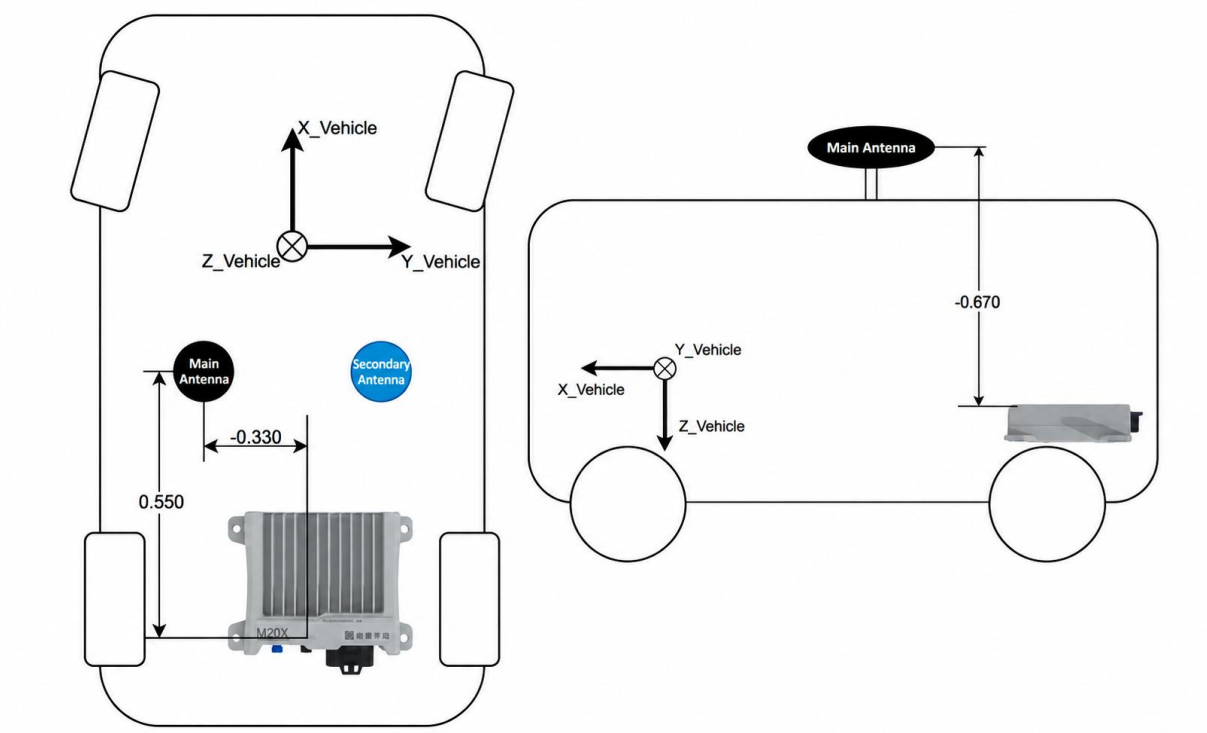


Figure 3-1. Lever arm from the receiver to the primary GNSS antenna

3.1.2.2 Rear-Axle Center Lever Arm Configuration

During normal vehicle motion, the lateral and vertical velocities at the rear-axle center are zero. To apply this constraint correctly and improve dead-reckoning performance, measure the lever arm from the receiver coordinate-system origin to the rear-axle center after installation. The measurement method is the same as for the GNSS antenna lever arm.

```
SETLEVERARM NHC X Y Z
```

X, Y, and Z are the rear-axle center lever-arm coordinates in metres. This parameter must be configured correctly when wheel-speed data is used; an incorrect value may cause abnormal navigation results.

3.1.2.3 Receiver Rotation Parameter Configuration

The attitude offset between the receiver coordinate system and the vehicle coordinate system is called the receiver rotation parameter, or installation angle. It is expressed as the Euler angles that rotate the vehicle coordinate system (Vehicle_Frame) into the receiver coordinate system (Body_Frame).

```
SETROTATIONANGLE X Y Z
```

X, Y, and Z are the rotation angles about the corresponding axes, in degrees. Apply the rotations in Z → Y → X order, using the right-hand rule.

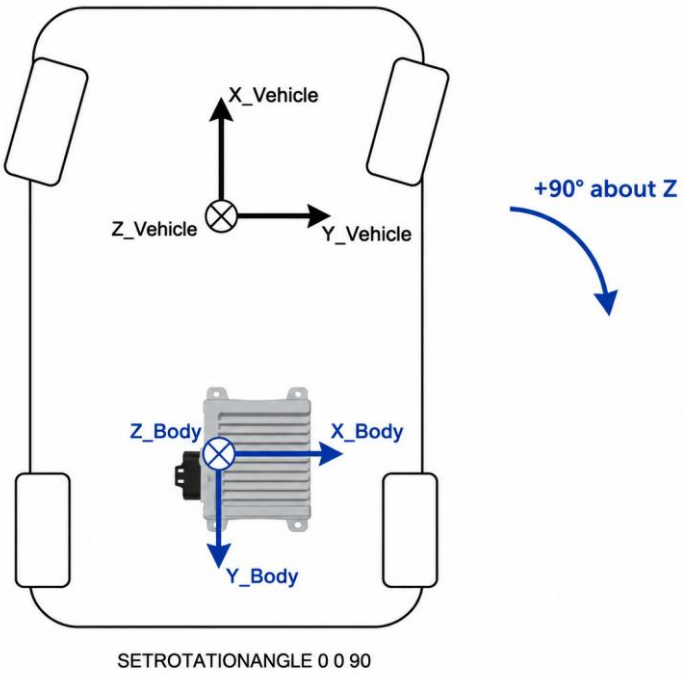


Figure 3-2. Relationship between the vehicle and receiver coordinate systems

For the installation shown in Figure 3-2, the vehicle coordinate system must be rotated +90° about the Z-axis to align with the receiver coordinate system. Configure the rotation as follows:

```
SETROTATIONANGLE 0 0 90
```

Note: Measure the rotation angles as accurately as possible. If the configured values contain errors, run online calibration.

3.1.2.4 Dual-Antenna Installation Angle Configuration

The M20X can use two GNSS antennas for heading determination, static initialization, and improved attitude accuracy. Before using dual-antenna heading, configure the installation angle according to the actual antenna arrangement.

Viewed from above, the dual-antenna vector points from the primary antenna to the secondary antenna. The installation angle is measured from the vehicle forward direction (the X-axis of the vehicle coordinate system) to this vector. Clockwise angles are positive and counterclockwise angles are negative, within

−180° to +180°. Verify that both antennas are connected to the correct receiver ports before configuration.

```
SETINSHEADINGDIFF PARAM
```

PARAM is the dual-antenna installation angle in degrees. Figure 3-3 shows the four common antenna arrangements; Table 3-2 lists their commands.

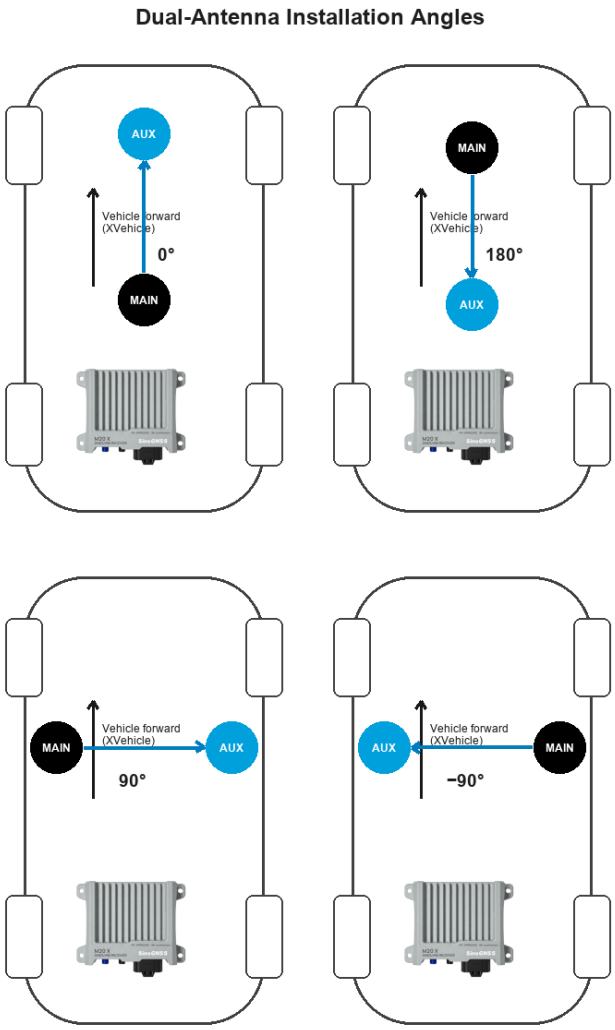


Figure 3-3. Common dual-antenna arrangements

Table 3-2. Dual-antenna installation commands

Primary / Secondary Arrangement	Angle	Command
Primary behind; secondary ahead	0°	SETINSHEADINGDIFF 0

Primary / Secondary Arrangement	Angle	Command
Primary ahead; secondary behind	180°	SETINSHEADINGDIFF 180
Primary left; secondary right	90°	SETINSHEADINGDIFF 90
Primary right; secondary left	−90°	SETINSHEADINGDIFF -90

3.1.3 Automotive Ethernet Configuration

Before connecting the M20X through Automotive Ethernet, configure the receiver's Ethernet port through a serial port. See Section 3.1.1 for serial communication. A computer cannot directly recognize a 100BASE-T1 Automotive Ethernet interface, so an appropriate Automotive-Ethernet-to-standard-Ethernet converter is required.

3.1.3.1 Static IP Address

Assign a unique IP address to both the receiver and the computer. Static addresses are normally used in a test environment. A USB-to-serial adapter is generally required to configure the receiver IP address; see the M20X Message Manual for command details.

Default receiver IP address: 192.168.1.104. Subnet mask: 255.255.255.0.

3.1.3.2 ICOM

ICOM is a virtual serial port for Automotive Ethernet communication and can use TCP or UDP. An ICOM interface can be configured as a TCP client, TCP server, or UDP endpoint. By default, the receiver uses TCP server mode on port 8089.

3.2 Operating Instructions

Before operating the M20X, complete the installation and configuration described in Chapters 2 and 3 and make sure that the receiver is powered on. To confirm operation through COM2, connect the port and send:

```
LOG VERSION
```

A normal response is similar to:

```
Receive command, [COM2]
<OK!
VERSION: xxx
```

If the command is invalid, the receiver returns an error. If no response is received, check the power supply, cable connection, selected serial port, and baud rate.

3.2.1 Differential Data Input

Differential corrections are sent from a reference station to the rover to improve rover position accuracy. The data normally use RTCM 3.x. A user may operate a private reference station or subscribe to a commercial service, but a communication link is always required. During testing, a computer and forwarding software such as STRSVR can provide the link; in field operation, a radio or DTU may be used. The M20X normally receives corrections through COM1 or Ethernet.

3.2.1.1 Serial-Port Input

1. Set the COM1 baud rate: **COM COM1 115200**

2. Save the configuration: **SAVECONFIG**

3. Connect COM1 to the correction-data source.

3.2.1.2 Automotive Ethernet Input

After completing the IP and ICOM configuration in Section 3.1.3, connect the receiver to the computer through the Ethernet converter. Create a connection in the network debugging tool and send the correction stream directly.

3.2.1.3 Checking RTK Status

After corrections are connected, monitor the position status through a serial or Ethernet interface.

For a serial connection, send:

```
LOG BESTPOSA ONTIME 1
```

For Ethernet, create another ICOM connection, such as ICOM2, and send:

```
LOG ICOM2 BESTPOSA ONTIME 1
```

In the BESTPOSA message, **NARROW_INT** indicates that valid differential data are being received and used.

3.2.2 Wheel-Speed Input

Wheel-speed aiding improves dead reckoning when GNSS signals are severely blocked or unavailable. The M20X normally receives wheel speed through CAN/CAN FD. Configure the rear-axle center lever arm before enabling wheel-speed input.

Wheel-speed update rate should be at least 20 Hz (50 Hz or higher recommended), accuracy better than 0.1 m/s (0.01 m/s recommended), resolution better than 0.02 m/s, and latency no greater than 10 ms. The receiver has no internal CAN termination resistor; install the required termination at the connection point.

After connection, verify wheel-speed reception with:

```
LOG RAWODOMA ONNEW
```

Continuous RAWODOMA messages indicate that the receiver is receiving wheel-speed data.

3.2.3 Real-Time Result Output

- Operate the receiver where GNSS signals can be received normally. With unavailable or poor-quality satellite signals, the output position, velocity, and attitude may not meet the application requirements.
- GNSS and INS navigation information can be obtained independently. The integrated solution is normally output in INS messages.

Non-INS messages support rates up to 5 Hz. INS messages support up to 100 Hz, or 200 Hz as an option. For example, output INSPVA in ASCII at 100 Hz with:

```
LOG INSPVAA ONTIME 0.01
```

To output the same message in binary format, use:

```
LOG INSPVAB ONTIME 0.01
```

For high-rate INS messages, use Ethernet and binary format to preserve data integrity. If a serial port is used, set its baud rate to 921600. Common output messages are listed in Table 3-3.

Table 3-3. Integrated navigation messages

Category	Messages
Position	INSPOS, INSPVA, INSPVAX
Velocity	INSVEL, INSSPD, INSPVA, INSPVAX

Category	Messages
Attitude	INSATT, INSPVA, INSPVAX
Solution confidence	INSSTDEV, INSPVAX

3.2.4 System Alignment

System alignment is the process by which the integrated navigation system obtains initial position, velocity, and attitude estimates. Monitor the INS_Status field in INSPVA or INSPVAX to follow alignment progress. The status definitions are summarized in Table 3-4.

1. After power-on, the system is inactive (**INS_INACTIVE**).
2. With good satellite reception, the system enters alignment mode (**INS_ALIGNING**) and uses GNSS data to determine initial position, velocity, and heading.
3. With dual antennas connected and the installation angle configured correctly, alignment can complete while stationary. Without valid dual-antenna heading, drive straight above the alignment speed threshold. The default is 3 m/s and can be changed with **SETALIGNVEL PARAM**.
4. After initialization, the status changes to **INS_ALIGNMENT_COMPLETE**.
5. GNSS data continue to refine the integrated solution. After sufficient turning manoeuvres and convergence, the status changes to **INS_SOLUTION_GOOD.**
6. If the integrated solution uncertainty becomes large, the status changes to **INS_HIGH_VARIANCE**. If GNSS navigation is unavailable, it changes to **INS_SOLUTION_FREE**.

Table 3-4. Inertial navigation status

Status	Description
INS_INACTIVE	INS is inactive.
INS_ALIGNING	INS is performing alignment.
INS_HIGH_VARIANCE	The INS solution may not meet the specified accuracy.
INS_SOLUTION_GOOD	INS is navigating with a converged, high-quality solution.
INS_SOLUTION_FREE	INS is navigating while GNSS is unavailable.

Status	Description
INS_ALIGNMENT_COMPLETE	Initialization is complete, but vehicle dynamics are not yet sufficient for full accuracy.

3.2.5 Online Calibration

After initial installation and configuration of the receiver rotation, GNSS antenna lever arm, and dual-antenna installation angle, online calibration can estimate more accurate values and improve integrated navigation accuracy. The calibrated parameters are the receiver rotation, GNSS antenna lever arm, and dual-antenna installation angle.

1. Power on the receiver and clear the previous calibration parameters: **CLEARDRCLBPARAM**
2. Restart the receiver with **RESET** and wait for the restart to complete.
3. Complete system alignment.
4. Drive for about 5 minutes at a steady speed above 5 m/s where possible, and never below 1 m/s. Use a figure-eight route, or a route containing both straight segments and turns. See Figures 3-4 and 3-5.
5. Start or stop calibration-status output with **LOG INSCALIBSTATE ONTIME 1** or **UNLOG INSCALIBSTATE.**
6. Monitor the INSCALIBSTATE message. It reports the current calibration run only; use READDRCLBPARAM to query the previously stored calibration state.
7. At every power-on, the receiver loads completed calibration states and applies the calibrated parameters automatically.

```
%INSCALIBSTATES,WEEK,GPST;GPST,STATE,LeverArmX,LeverArmY,LeverArmZ,MisAngleX,MisAngleY,MisAngleZ,HeadingDiff
```

The STATE field is a bit mask. Completed parameter states are added together; for example, 0x07 indicates completion of lever-arm, receiver-installation-angle, and dual-antenna-angle calibration. The completion bits are listed in Table 3-5.

Table 3-5. Online calibration completion bits

Calibration Parameter	Completion Bit
GNSS antenna lever arm	0x01
Receiver installation angle	0x02

Calibration Parameter	Completion Bit
Dual-antenna installation angle	0x04



Figure 3-4. Recommended online-calibration route 1



Figure 3-5. Recommended online-calibration route 2

3.2.6 Data Post-Processing

The raw data output by the M20X can be processed with professional post-processing software. Bidirectional processing and smoothing can produce a more accurate reference trajectory for test evaluation.

3.2.7 Firmware Upgrade

Firmware may need to be upgraded as the product is improved. Download the firmware package from the company website and follow the separate Firmware Upgrade Instructions.

4 Appendix A: Message Descriptions

4.1 INSPVA — INS Position, Velocity, and Attitude

Example:

```
#INSPVAA,USB1,0,67.5,FINESTEERING,2209,490558.000,02000020,18bc,16809;2209,490558.000000000,51
.15043714042,-114.03067871718,1080.3548,0.0051,-0.0014,-0.0012,-
0.296402993,0.311887972,157.992156267,SOLUTION_GOOD*cc698020
```

Table 4-1. INSPVA fields

No.	Field	Description	Type	Bytes	Binary Offset
1	Header	Message header	—	H	0
2	Week	GPS week (GPS time)	Ulong	4	H
3	Seconds	Seconds of GPS week	Double	8	H+4
4	Latitude	Latitude, WGS 84 [deg]	Double	8	H+12
5	Longitude	Longitude, WGS 84 [deg]	Double	8	H+20
6	Height	Ellipsoidal height [m]	Double	8	H+28
7	North Velocity	North velocity; negative is south [m/s]	Double	8	H+36
8	East Velocity	East velocity; negative is west [m/s]	Double	8	H+44
9	Up Velocity	Up velocity [m/s]	Double	8	H+52
10	Roll	Rotation about X, right-hand positive [deg]	Double	8	H+60
11	Pitch	Rotation about Y, right-hand positive [deg]	Double	8	H+68
12	Azimuth	Rotation about Z, right-hand positive, 0–360° [deg]	Double	8	H+76

No.	Field	Description	Type	Bytes	Binary Offset
13	Status	INS status	Enum	4	H+84
14	xxxx	32-bit CRC	Hex	4	H+88
15	[CR][LF]	ASCII terminator	—	—	—

Table 4-2. INS solution status codes

Binary	Status	Description
0	INACTIVE	Alignment has not started.
1	ALIGNING	INS is aligning.
2	HIGH_VARIANCE	The INS solution may not meet accuracy requirements.
3	SOLUTION_GOOD	INS is navigating with a good solution.
6	SOLUTION_FREE	INS is navigating while GNSS is unavailable.
7	ALIGNMENT_COMPLETE	INS is navigating, but vehicle dynamics are insufficient for full accuracy.

4.2 INSATT — INS Attitude

Example:

```
#INSATTA,USB2,0,14.5,FINESTEERING,1541,487970.000,02040000,5b35,37343;1541,487970.000549050,1.876133508,4.053672765,328.401460897,SOLUTION_GOOD*ce4ac533
```

Table 4-3. INSATT fields

No.	Field	Description	Type	Bytes	Binary Offset
1	Header	Message header	—	H	0
2	Week	GPS week	Ulong	4	H
3	Seconds	Seconds of GPS week	Double	8	H+4
4	Roll	Rotation about X, right-hand positive [deg]	Double	8	H+12
5	Pitch	Rotation about Y, right-hand positive [deg]	Double	8	H+20
6	Azimuth	Clockwise angle from true north [deg]	Double	8	H+28
7	Status	INS status	Enum	4	H+36
8	xxxx	32-bit CRC	Hex	4	H+40
9	[CR][LF]	ASCII terminator	—	—	—

4.3 INSSPD — INS Horizontal and Vertical Speed

Example:

```
#INSSPDA,USB2,0,20.0,FINESTEERING,1541,487969.000,02040000,7832,37343;1541,487969.000549050,32
9.621116190,14.182070674,-0.126606551,SOLUTION_GOOD*c274fff2
```

Table 4-4. INSSPD fields

No.	Field	Description	Type	Bytes	Binary Offset
1	Header	Message header	—	H	0
2	Week	GPS week	Ulong	4	H

No.	Field	Description	Type	Bytes	Binary Offset
3	Seconds	Seconds of GPS week	Double	8	H+4
4	Trk gnd	Direction of motion over ground from true north [deg]	Double	8	H+12
5	Hor spd	Horizontal speed [m/s]	Double	8	H+20
6	Vert spd	Vertical speed; positive upward [m/s]	Double	8	H+28
7	Status	INS status	Enum	4	H+36
8	xxxx	32-bit CRC	Hex	4	H+40
9	[CR][LF]	ASCII terminator	—	—	—

4.4 INSPOS — INS Position

Example:

```
#INSPOSA,USB2,0,18.0,FINESTEERING,1541,487977.000,02040000,17cd,37343;1541,487977.000549050,51
.121315135,-114.042311349,1038.660737046,SOLUTION_GOOD*2fffd557
```

Table 4-5. INSPOS fields

No.	Field	Description	Type	Bytes	Binary Offset
1	Header	Message header	—	H	0
2	Week	GPS week	Ulong	4	H
3	Seconds	Seconds of GPS week	Double	8	H+4
4	Latitude	Latitude, WGS 84 [deg]	Double	8	H+12
5	Longitude	Longitude, WGS 84 [deg]	Double	8	H+20

No.	Field	Description	Type	Bytes	Binary Offset
6	Height	Ellipsoidal height [m]	Double	8	H+28
7	Status	INS status	Enum	4	H+36
8	xxxx	32-bit CRC	Hex	4	H+40
9	[CR][LF]	ASCII terminator	—	—	—

4.5 INSSTDEV — INS PVA Standard Deviations

Example:

```
#INSSTDEVA,USB1,0,66.0,FINESTEERING,2209,491004.000,02000020,2396,16809;0.1813,0.1813,0.1806,0
.0018,0.0018,0.0017,0.0292,0.0291,0.0577,13000045,0,0,7fd1bf,0*b490ddee
```

Table 4-6. INSSTDEV fields

No.	Field	Description	Type	Bytes	Binary Offset
1	Header	Message header	—	H	0
2	Latitude σ	Latitude standard deviation [m]	Float	4	H
3	Longitude σ	Longitude standard deviation [m]	Float	4	H+4
4	Height σ	Height standard deviation [m]	Float	4	H+8
5	North Velocity σ	North-velocity standard deviation [m/s]	Float	4	H+12
6	East Velocity σ	East-velocity standard deviation [m/s]	Float	4	H+16
7	Up Velocity σ	Up-velocity standard deviation [m/s]	Float	4	H+20
8	Roll σ	Roll standard deviation [deg]	Float	4	H+24

No.	Field	Description	Type	Bytes	Binary Offset
9	Pitch σ	Pitch standard deviation [deg]	Float	4	H+28
10	Azimuth σ	Azimuth standard deviation [deg]	Float	4	H+32
11	Ext sol stat	Extended solution status	Ulong	4	H+36
12	Time Since Update	Time since last ZUPT or position update [s]	Ushort	2	H+40
13	Reserved	Reserved	Ushort	2	H+42
14	Reserved	Reserved	Ulong	4	H+44
15	Reserved	Reserved	Ulong	4	H+48
16	xxxx	32-bit CRC	Hex	4	H+52
17	[CR][LF]	ASCII terminator	—	—	—

4.6 INSVEL — INS North, East, and Up Velocity

Example:

```
#INSVELA,USB1,0,19.0,FINESTEERING,1543,236173.000,02000000,9c95,37343;1543,236173.002500000,14
.139471871,-0.070354464,0.044204369,SOLUTION_GOOD*3c37c0fc
```

Table 4-7. INSVEL fields

No.	Field	Description	Type	Bytes	Binary Offset
1	Header	Message header	—	H	0
2	Week	GPS week	Ulong	4	H
3	Seconds	Seconds of GPS week	Double	8	H+4

No.	Field	Description	Type	Bytes	Binary Offset
4	North Velocity	North velocity [m/s]	Double	8	H+12
5	East Velocity	East velocity [m/s]	Double	8	H+20
6	Up Velocity	Up velocity [m/s]	Double	8	H+28
7	Status	INS status	Enum	4	H+36
8	xxxx	32-bit CRC	Hex	4	H+40
9	[CR][LF]	ASCII terminator	—	—	—

4.7 RAWIMU — Raw Inertial Measurements

Example:

```
#RAWIMUA,COM2,0,57.0,FINESTEERING,2004,28212.750,00000000,000e,6480;2004,28212.750,00000000,433,-19,-114,1,-16,-20*67518841
```

Table 4-8. RAWIMU fields

No.	Field	Description	Type	Bytes	Binary Offset
1	Header	Message header	—	H	0
2	Week	GPS week	Ulong	4	H
3	Seconds	Seconds of GPS week	Double	8	H+4
4	IMU Status	1 = valid, 2 = invalid; current default is 0	Hex	4	H+12
5	Z Accel Output	Z-axis acceleration; scale factor depends on IMU	Long	4	H+16

No.	Field	Description	Type	Bytes	Binary Offset
6	−Y Accel Output	Negative Y-axis acceleration; scale factor depends on IMU	Long	4	H+20
7	X Accel Output	X-axis acceleration; scale factor depends on IMU	Long	4	H+24
8	Z Gyro Output	Z-axis angular-rate output, right-hand positive	Long	4	H+28
9	−Y Gyro Output	Negative Y-axis angular-rate output, right-hand positive	Long	4	H+32
10	X Gyro Output	X-axis angular-rate output, right-hand positive	Long	4	H+36
11	xxxx	32-bit CRC	Hex	4	H+40
12	[CR][LF]	ASCII terminator	—	—	—

5 Appendix B: Common Coordinate Systems

5.1 Coordinate Systems Used in Integrated Navigation

The coordinate systems commonly used in integrated navigation are the local navigation frame, receiver frame, vehicle frame, and user-defined frame.

5.1.1 Local Navigation Frame

The local navigation frame is the N-E-D (North-East-Down) frame:

• N-axis: points north in the plane perpendicular to the D-axis.

• E-axis: points east and completes the right-handed orthogonal frame with N and D.

• D-axis: points down along the normal to the reference ellipsoid.

The origin is the navigation center marked on the receiver enclosure. See Figure 5-1.

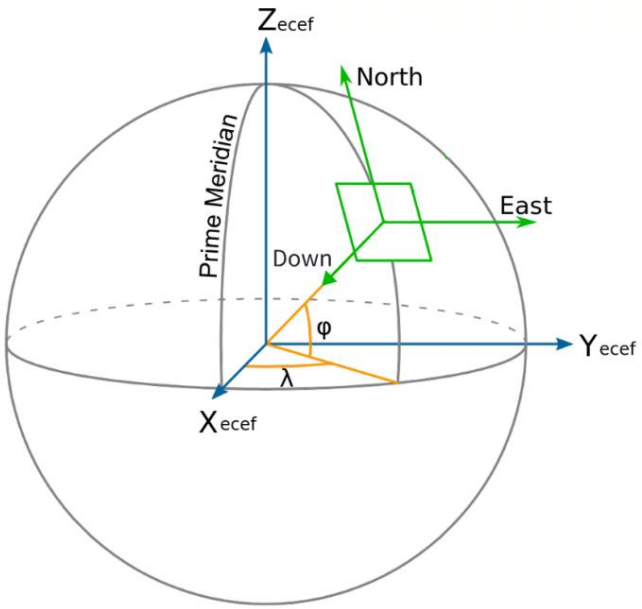


Figure 5-1. Local navigation frame

5.1.2 Receiver Coordinate System

The receiver coordinate-system origin and axis directions are marked on the receiver enclosure. The marked navigation center is the origin. See Figure 5-2.



Figure 5-2. Receiver coordinate system

5.1.3 Vehicle Coordinate System

The origin of the vehicle coordinate system is the navigation center marked on the receiver enclosure. Its axes are defined as follows:

• Z-axis: perpendicular to the vehicle floor and pointing downward.

• X-axis: pointing in the vehicle's forward direction.

• Y-axis: completing the right-handed orthogonal frame defined by X and Z.

When using the land-vehicle model, configure antenna lever arms and rotation parameters strictly according to this vehicle coordinate system. See Figure 5-3.

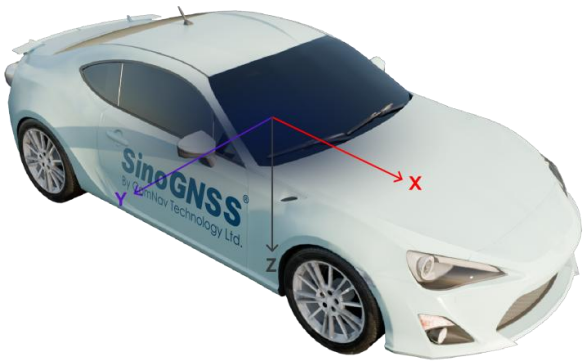


Figure 5-3. Vehicle coordinate system

5.1.4 User-Defined Coordinate System

A user-defined coordinate system is a right-handed frame whose origin and axis directions are specified by the user. By default, the receiver outputs position and velocity at the navigation center. Use SETLEVERARM USER X Y Z to apply a known coordinate offset and move the output origin to another point.

6 Troubleshooting

If one of the following problems occurs, use the listed checks. Contact ComNav Technology technical support if the problem remains.

6.1 Receiver Does Not Start

1. Check that the supply voltage is within the specified range. Disconnect and replace a faulty power source before using the product.
2. Check that the power cable is connected correctly and securely.
3. Measure the connector voltage with a multimeter and inspect the cable. Replace a damaged cable.
4. If these checks do not identify the cause, the receiver may be damaged; contact technical support.

6.2 No Response from the Serial Port

1. Verify the selected COM port and baud rate against the communication protocol.
2. Inspect the data cable for damage.
3. Verify that the required serial-port driver is installed.
4. Check that the data cable is connected securely.

6.3 Abnormal Positioning

1. Check whether the receiver or GNSS antennas are obstructed.
2. Verify connectivity of all communication equipment.
3. Check for RF interference within the GNSS antenna frequency bands.
4. Confirm that INS initialization has completed successfully.
5. Confirm that the reference-station coordinates do not differ significantly from their true values.

7 Important Notes

1. Do not disassemble any part of the system. If a fault occurs, record the symptoms and contact technical support.
2. Observe the operating-voltage requirements of every device. Use the supplied power adapter and data cable to prevent damage.
3. Do not reverse the positive and negative power connections.
4. Install and wire the equipment exactly as described in this manual, and make sure every connector is fully seated.
5. Stop using damaged cables or accessories and replace them promptly.